

Competency Based Learning Material



Sector:	AGRICULTURE, FORESTRY AND FISHERY
Qualification :	ORGANIC AGRICULTURE PRODUCTION NC II
Unit of Competency:	PRODUCE ORGANIC FERTILIZER
Module Title	PRODUCING ORGANIC FERTILIZER
MDM SAGAY COLLEGE INC.	

ORGANIC AGRICULTURE PRODUCTION NC II

The **ORGANIC AGRICULTURE PRODUCTION NC II** Qualification consists of competencies that a person must achieve to produce organic farm products such as chicken and vegetables including producing of organic supplements such as fertilizer, concoctions and extracts. It has two (2) elective competencies which are on raising organic hogs and raising organic small ruminants.

This Qualification is packaged from the competency map of the Agri-Fishery Sector.

The units of competency comprising this qualification includes the following:

CODE	BASIC COMPETENCIES
500311105	Participate in workplace communication
500311106	Work in a team environment
500311107	Practice career professionalism
500311108	Practice occupational health and safety procedures

CODE	COMMON COMPETENCIES
AFF 321201	Apply safety measures in farm operations
AFF 321202	Use farm tools and equipment
AFF 321203	Perform estimation and calculations
TRS311201	Develop and update industry knowledge
AGR321205	Perform record keeping

CODE	CORE COMPETENCIES
AGR612301	Raise organic chicken
AGR611306	Produce organic vegetables
AGR611301	Produce organic fertilizer
AGR611302	Produce organic concoctions and extract
CODE	ELECTIVE COMPETENCIES
AGR612302	Raise organic hogs
AGR612303	Raise organic small ruminants

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 2
			Revision # 00	

HOW TO USE THIS MODULE

Welcome to the Module on Producing Organic Fertilizer. This module contains training materials and activities for you to complete.

The unit of competency on *Produce organic fertilizer* contains knowledge, skills and attitudes required for an Organic Agriculture NC II course.

You are required to go through a series of learning activities in order to complete each of the learning outcomes of the module. In each learning outcome there are **Information Sheets**, **Operation Sheets**, **Job Sheets** and **Task Sheets**. Follow these activities on your own and answer the Self-Check at the end of each learning activity.

If you have questions, don't hesitate to ask your trainer for assistance.

Recognition of Prior Learning (RPL)

You may already have some of the knowledge and skills covered in this module because you have:

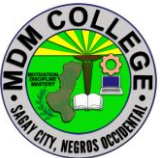
- been working for some time
- already have completed training in this area.

If you can demonstrate to your trainer/instructor that you are competent in a particular skill or skills, talk to him/her about having them formally recognized so you don't have to do the same training again. If you have a qualification or Certificate of Competency from previous trainings show it to your trainer. If the skills you acquired are still current and relevant to this module, they may become part of the evidence you can present for RPL. If you are not sure about the currency of your skills, discuss it with your trainer.

After completing this module ask your trainer to assess your competency. Result of your assessment will be recorded in your competency profile. All the learning activities are designed for you to complete at your own pace.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 3
			Revision # 00	

Inside this module you will find the activities for you to complete followed by relevant information sheets for each learning outcome. Each learning outcome may have more than one learning activity.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 4
			Revision # 00	

MODULE CONTENT

Program/Course : Organic Agriculture) NC II
Unit of Competency : Produce Organic Fertilizer
Module : Producing Organic Fertilizer

INTRODUCTION:

This module contains information and suggested learning activities on producing organic fertilizer. It includes activities and materials on Organic Agriculture Production NC II

Completion of this module will help you better understand the succeeding module on the raising organic chicken.

This module consists of 2 learning outcomes. Each learning outcome contains learning activities supported by each instruction sheets. Before you perform the instructions, read the information sheets and answer the self-check and activities provided to ascertain to yourself and your trainer that you have acquired the knowledge necessary to perform the skill portion of the particular learning outcome.

Upon completion of this module, report to your trainer for assessment to check your achievement of knowledge and skills requirement of this module. If you pass the assessment, you will be given a certificate of completion.

SUMMARY OF LEARNING OUTCOMES:

Upon completion of the module you should be able to:

- LO1 Prepare composting area and raw materials
- LO2 Compost and harvest fertilizer

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 5
			Revision # 00	

Learning Outcome #1 Prepare composting area and raw materials

Assessment Criteria:

Site is selected based on compost fertilizer production requirements and

Site lay-out is prepared based on location.

Bed is prepared in accordance with production requirements

Materials are gathered based on production requirements and PNS for organic fertilizer

Prepare raw materials following enterprise procedure and PNS for organic fertilizer

Contents:

- Site Selection
- Prepare site layout
- Prepare bed
- Gather material
- Prepare raw materials

Tools, Materials and Equipment and Facilities

- Composting shed using locally available materials
- Shredder
- Drums for water storage
- Flat and pointed shovels
- Hoe
- Wheelbarrow
- Sieve
- Water hose

References:

<https://www.progressivegardening.com/waste-management/siting-and-area-considerations.html>

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 6
			Revision # 00	

LEARNING EXPERIENCES/ACTIVITIES

LO # 1 Prepare composting area and raw materials

Learning Activities	Special Instructions
Read the attached Information Sheet 3.1-1 on Site Selection	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.1-1 on Site Selection	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.1-1 on Site Selection	
Perform Job Sheet 3.1-1 on Site Selection	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 3.1-1 on Site Selection	
Read the attached Information Sheet 3.1-2 on Prepare site layout	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.1-2 on Prepare site layout	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.1-2 on Prepare site layout	
Perform Job Sheet 3.1-2 on Prepare site layout	Observe the demonstration of your trainer and check the procedures while performing the operation.

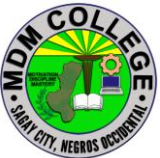
	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 7
			Revision # 00	

Evaluate performance using performance checklist on 3.1-2 on Prepare site layout	
Read the attached Information Sheet 3.1-3 on Prepare bed	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.1-3 on Prepare bed	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.1-3 on Prepare bed	
Perform Job Sheet 3.1-3 on Prepare bed	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 2.1-3 on Prepare bed	
Read the attached Information Sheet 3.1-4 on Gather material	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.1-4 on Gather material	Try to answer the Self-Check without looking at the information sheet
Perform Job Sheet 3.1-4 on Gather material	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 3.1-4 on Gather material	

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 8
			Revision # 00	

Read the attached Information Sheet 3.1-5 on	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.1-5 on Prepare raw materials	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.1-5 on Prepare raw materials	
Perform Job Sheet 3.1-5 on Prepare raw materials	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 3.1-5 on Prepare raw materials	

After doing this activity, you are now ready to proceed to the next module on Compost and harvest fertilizer.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 9
			Revision # 00	

INFORMATION SHEET 3.1-1

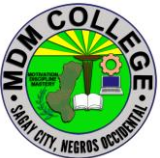
Site Selection

Learning Objective: After reading this information sheet, you should be able to select composting site.

Introduction:

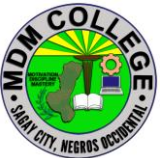
The location of the composting facility is a very important factor in a successful compost operation. To minimize material handling, the composting facility should be located as close as possible to the source of organic waste. If land application is the preferred method of utilization, the facility should also be located with convenient access to the land application sites. Several other important considerations when locating a compost facility are discussed below.

1. Wind direction—improperly managed compost facilities may generate offensive odors until corrective actions are taken. Wind direction and proximity to neighbors should be considered when locating a composting facility.
2. Topography—Avoid locating composting facilities on steep slopes where runoff may be a problem and in areas where the composting facility will be subject to inundation.
3. Ground water protection - the composting facility should be located downgradient and at a safe distance from any wellhead. A roofed compost facility, that is properly managed, should not generate leachate that could contaminate ground water. If a compost facility is not protected from the weather, it should be sited to minimize the risk to ground water.
4. Area requirements—the area requirements for each composting method vary. The windrow method requires the most land area. The static pile method requires less land area than the windrow method, but more than the in-vessel method. The pile dimensions also affect the amount of land area necessary for composting. A large pile that has a low surface area to total volume ratio requires less composting area for a given volume of manure, but it is also harder to manage. The size and

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 10
			Revision # 00	

type equipment used to mix, load, and turn the compost should also be considered when sizing a compost area. Enough room must be provided in and around the composting facility to operate equipment. In addition, a buffer area around the compost site should be considered if a visual barrier is needed or desired. In general, given the pile dimensions, a compost bulk density of 35 to 45 pounds per cubic feet can be used to estimate the surface area necessary for stacking the initial compost mix. To this area, add the amount of area necessary for equipment operation, pile turning, and buffer.

5. Existing areas—to reduce the initial capital cost, existing roofed, concrete, paved, or gravel areas should be used if possible as a composting site.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 11
			Revision # 00	

SELF CHECK 3.1-1

Site Selection

Identification

1. _____. Wind direction and proximity to neighbors should be considered when locating a composting facility.
2. _____ Avoid locating composting facilities on steep slopes where runoff may be a problem
3. _____ A roofed compost facility, that is properly managed, should not generate leachate that could contaminate ground water
4. _____ Requirements for each composting method vary. The windrow method requires the most land area.
5. _____ reduce the initial capital cost, existing roofed, concrete, paved, or gravel areas should be used.

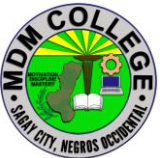
	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 12
			Revision # 00	

ANSWER KEY 3.1-1

Site Selection

Identification

1. Wind direction
2. Topography
3. Ground water protection
4. Area requirements
5. Existing areas

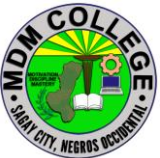
	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 13
			Revision # 00	

JOB SHEET 3.1-1

Site Selection

Title: Select composting Site
Performance Objective: At the end of this module the student should be able to select composting site.
Supplies/Materials: PPE, composting area,
Steps/Procedures: Wear the PPE Observe the composting area Select the area which is not sloppy Do housekeeping
Assessment Method: Demonstration

PERFORMANCE CHECKLIST 3.1-1

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 14
			Revision # 00	

Site Selection

Performance Criteria	YES	NO
Did the trainee...		
Wear the PPE? Wear the PPE?		
Observe the composting area?		
Select the area which is not sloppy?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 15
			Revision # 00	

INFORMATION SHEET 3.1-2

Prepare site layout

Learning Objective: After reading this information sheet, you should be able to prepare composting site layout.

Introduction:

A composting facility is a structure or device that uses controlled aerobic decomposition to transform waste organic material into a biologically stable product that can be used as a soil amendment.

Layout determines how efficiently material is handled and can significantly impact the quality of the end products. A good layout minimizes material handling and helps control outside factors, like moisture, that can impact maturing and cause odors. And while the overview of how a composting facility is constructed can seem pretty basic, it's the little details that matter.



1: Drop-off Area

How and where incoming organic waste is accepted is a critical detail. Because composting facilities typically charge tip fees, the first thought is how to make tipping more convenient for people to dump and leave. So they put the drop-off area just beyond the weigh station. And if the site has a retail yard to sell compost and/or mulch, it also wants to have that area near the entrance of the facility. Having both raw feedstocks and finished products up front requires a lot of

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 16
			Revision # 00	

movement, and extra time spent moving material can have a significant impact on fixed operating expenses, reducing profit margins.

The more efficient place to accept incoming material is closer to the grinding area, so as it's processed, it can gradually move toward the front and be ready for retail. Controlling the flow from where the material is received can significantly reduce the number of times it's moved with loaders, helping to save on equipment costs, fuel and labor. The other advantage of having the drop-off area away from the front of the facility is to reduce the possibility of incoming material accidentally getting mixed in with finished material waiting to be sold.

2: Incoming Material and Grinding Area

Separate from the drop-off area, but right alongside it, is where stacked raw material is staged to be run through a grinder. Only trained employees should have access to this area of the operation.

What is important at Stop #2 is material flow and efficient handling. Grinding is one of the highest cost functions of any operation so it's essential to fully use the grinder throughput potential. Materials should be able to be pushed from the drop-off area to the grinder, and the discharge should be pointed in the direction of the composting area.

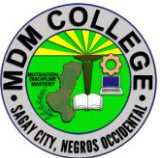
3: Composting Area

In the composting area, look at operations from the ground up, starting with the base and then looking at how the area drains storm water runoff. A dirt pad may be the most economical option, but it's not going to be the most efficient or create a higher quality compost. For example, when dealing with wet material, like food waste, mulch and rain, dirt pads can quickly get sloppy and potentially give off an odor. Concrete or asphalt pads often offer the best surface for quality compost production.

Another critical detail is to optimize drainage in the composting area by having a slope so there are places for water to go after a heavy rain, for example. To keep water from becoming trapped between compost rows, windrows should run parallel with the slope of the pad. Spacing of windrows should be kept to a minimum to help maximize usage of the composting pad.

4: Screening and Finishing Area

Finished product screening should be positioned between the composting area and retail space to help minimize the material handling involved. Position trommel screen so overs are coming off the conveyor near the composting side of the yard

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 17
			Revision # 00	

and fines are near the retail space. Simple steps like this help reduce cross contamination of material, which in turn results in a higher quality end product.

5: Retail Area

Setup and layout of the retail space depends on the customer base. If you're selling in bulk, piles are okay. If products are being bagged, position bulk material as close as possible to help minimize cycle times. Barriers should be put in place to minimize contaminants, like plastic and paper, from blowing into the retail area. Drainage and coverage should also be considered to keep materials dry and to maintain good quality.

6: Drainage Pond and Perimeter

Constructing a series of drainage ponds on the low side of the property near the composting rows serves to catch runoff and clean it. The first pond collects runoff while each additional pond helps slow the flow and give sediments a chance to separate. These ponds are also a good resource when the compost's moisture levels are low. Running a pump and sprinkler is a cost-effective way to bring moisture content back to optimal levels.

Establishing a barrier of trees or fencing to catch light plastic contaminants can be a good idea and is often required by state and local officials. Fencing is almost always required, but trees can offer an additional barrier of protection.

As your operation grows and evolves, the site layout and flow of material needs to be questioned and changes will likely be necessary. Operating on the basis of continuous improvement will guide you to a more efficient composting facility layout.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 18
			Revision # 00	

SELF CHECK 3.1-2

Prepare site layout

TRUE or FALLS

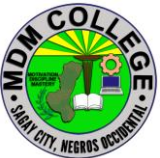
____1. A good layout minimizes material handling and helps control outside factors, like moisture, that can impact maturing and cause odors.

____2. The more efficient place to accept incoming material is closer to the grinding area, so as it's processed, it can gradually move toward the front and be ready for retail.

____3. Barriers should be put in place to minimize contaminates, like plastic and paper, from blowing into the retail area.

____4. Running a pump and sprinkler is a cost-effective way to bring moisture content back to optimal levels.

____5. Materials should be able to be pushed from the drop-off area to the grinder, and the discharge should be pointed in the direction of the composting area.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 19
			Revision # 00	

ANSWER KEY 3.1-2

Prepare site layout

1. TRUE
2. TRUE
3. TRUE
4. TRUE
5. TRUE

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 20
			Revision # 00	

JOB SHEET 3.1-2

Prepare site layout

Title: Preparing site layout	
Performance Objective: At the end of this module the student should be able to perform and preparing composting site layout.	
Supplies/Materials: PPE, composting area, Tools and Materials in layouting	
Steps/Procedures: Wear the PPE Determine good layout minimizes material handling and helps control outside factor Barriers should be put in place to minimize contamination. Running a pump and sprinkler is a cost-effective way to bring moisture content back to optimal levels Do housekeeping	

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 21
			Revision # 00	

PERFORMANCE CHECKLIST 3.1-2

Prepare site layout

Performance Criteria	YES	NO
Did the trainee...		
Wear the PPE?		
Determine good layout minimizes material handling and helps control outside factor?		
Barriers should be put in place to minimize contamination?		
Running a pump and sprinkler is a cost-effective way to bring moisture content back to optimal levels?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 22
			Revision # 00	

INFORMATION SHEET 3.1-3

Prepare bed

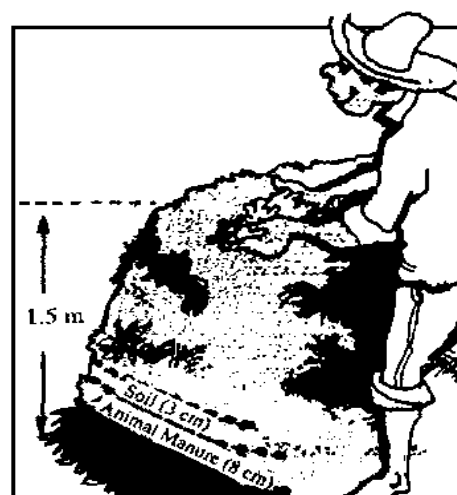
Learning Objective: After reading this information sheet, you should be able to prepare bed for composting.

Introduction:

Composting is a straight forward process that can be accomplished by anyone with a small patch of outdoor space. Composting allows you to divert your kitchen scraps and yard waste from a landfill, instead of turning them into a rich soil amendment. You only need a few tools, a little bit of time, and some basic information to begin composting and ensure the success of your setup. This guide will help you to create a compost bed (or compost "pile") from scratch and maintain it properly.

Method of compost preparation

1. Choose a spot that is at least partially protected from rain.
2. Gather the crop residues, animal manures and other wastes and bring them to the preparation site.
3. Pile the crop and other plant residues (15 cm thick) first. For the next layer, spread the animal manure to a thickness of about 8 cm, followed by about 3 cm of good soil. Pile another layer of the materials in the same sequence and repeat until a height of about 1.5 meters of the compost pile is attained.

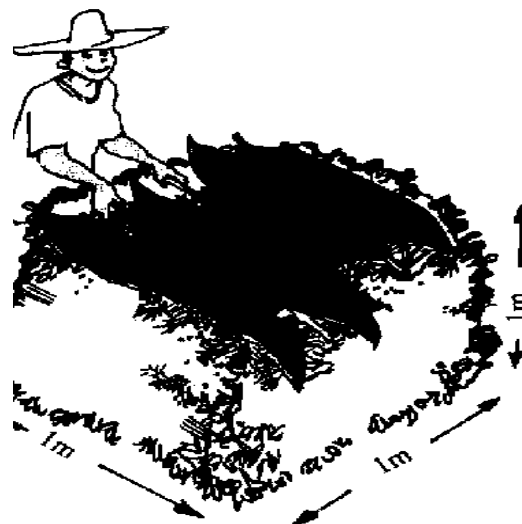


4. Water the pile until it is sufficiently moist. Water regularly.
5. Turn over or mix the pile with spading fork after 3 weeks, then again after five weeks.
6. Harvest the compost in three to four months.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 23
			Revision # 00	

The 14-day method of composting

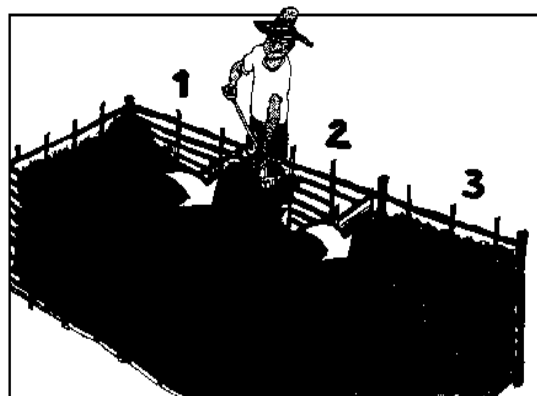
1. Chop the vegetative materials/plant wastes (dry or green or both).
2. Thoroughly mix these with an equal amount of fresh manure.
3. Pile the mixture into a heap measuring at least 1 m × 1 m × 1 m. (However, 1 m is the maximum height.)
4. Cover the heap with banana leaves or damaged burlap sacks.
5. By the third or fourth day, the inside of the heap should be heated up. If not, mix more manure into it.
6. On the same day (3rd or 4th), turn the heap inside out so that the materials from the center will appear outside and vice versa.
7. Turn the heap every two days thereafter.
8. In 14 - 18 days, the compost should be ready for use.



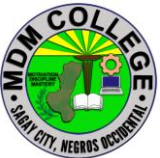
Composting in triple-compost bin

Making three compartments permits us to keep adding to our compost pile. The compartment at left is ready for the fields while the others are still rotting.

1. Fill compartment one with composting materials.
2. Add a small amount of soil or animal manure.
3. Continue in this way till the compartment is full.



4. After a month, empty the contents of compartment one into compartment two, mixing, watering and breaking up the compost in the process

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 24
			Revision # 00	

5. Cover the second compartment with a layer of soil, which has to be kept humid and loose.
6. Once compartment one is empty, the process of filling it should begin again as before.
7. After another month, fill compartment three with the contents of compartment two, airing the contents well without turning over.
8. Cover the third compartment with a layer of soil.
9. Fill compartment two with the contents of compartment one and cover with soil.
10. Fill compartment one with refuse and the cycle goes on.

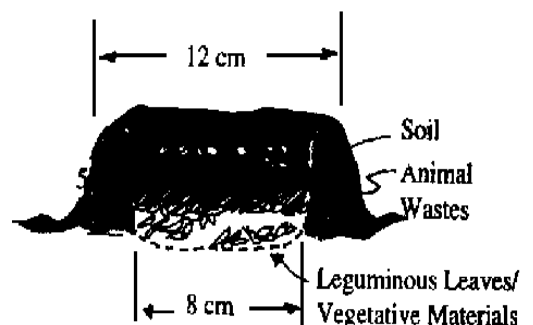
Deep bed composting



Cross-section of Composting Bed

Lay out garden beds at least 12 cm wide.

Dig trench 8 cm wide and 5 cm deep along center line of bed. Place spoil (dirt from trench) on both sides of trench.



	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 25
			Revision # 00	

Bed Construction

Addition of Organic Materials

Place 15 - 30 cm layer of leguminous leaves and other vegetative materials.

Spread layer of animal wastes over vegetative materials.

Cover with layer of soil. Use $\frac{1}{2}$ of spoil pile alongside of trench.

Pile another layer of the materials in the same sequence, returning all of spoils in or on trench. Shape bed by raking.



Planting

Soak bed thoroughly with water.

Plant seeds or transplant seedlings around the trench.

After harvesting, remove the contents of the trench and work the compost into the soil around the trench. Place new compost materials in the trench for the next crop.

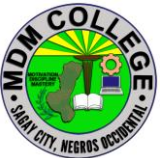
Semi-sunken composting

1. Clean the area selected for building the compost pile. Dig a hole one-half meter deep.

Clean the area

2. Cut composting materials into small pieces. Mix them with manure at 5:1 ratio.



	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 26
			Revision # 00	

Cut composting materials into small pieces

3. Place the mixture in the hole until it reaches one to two meters above the ground. Use a shovel or your hands to keep the edges square.

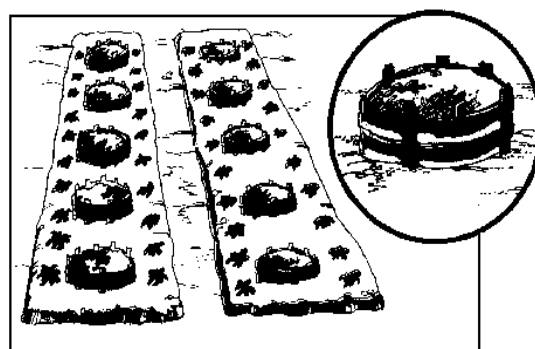
Place the mixture in the hole

4. Cover the pile with straw or smear it with mud to protect it. Add a layer of soil on top of the pile and make a series of holes on top of the finished pile. The compost should be ready in 1 to 2 months.



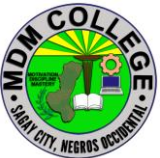
Basket composting

Basket composting is the process by which decomposable home garbage, garden and farm waste and leguminous leaves are allowed to rot in baskets half-buried in garden plots as a method of producing organic fertilizer.



Benefits

1. Basket compost can be used immediately without waiting for the usual 34 month period as is necessary in other methods of composting.
2. Baskets hold the composting materials in place, hence minimizing nutrient depletion by runoff.
3. Stray animals and fowls are prevented from scattering the compost materials.
4. Since garbage and wastes are collected and utilized, home and surroundings will become cleaner.

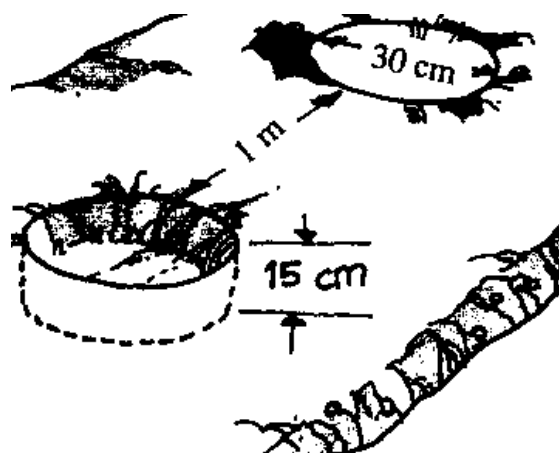
	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	DOCUMENT NO.	
		Nonie Falcunit	Issued by:	Page 27
			Revision # 00	

5. It serves as reservoir and collector of the moisture and nutrients.

6. More nutritious vegetables can be produced at less cost.

Preparation of Materials:

Long bamboo strips, 2-3 cm in width.
Bamboo stakes at least 30 cm in length.
Home organic garbage, farm and garden wastes, leguminous leaves.
Manure.



Preparation of Garden Plots

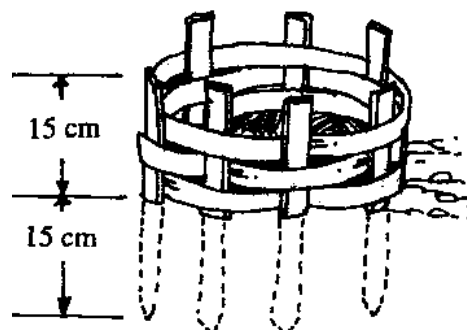
Clean garden site, save weeds and grasses for composting.
Dig at least 30 cm deep and raise the bed.
Dig holes along the center of the plots at least 15 cm in depth and 30 cm in diameter.
Space them 1 m apart

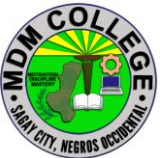
Preparation of Garden Plots

Construction of Baskets

Drive 7 stakes around the holes; uneven number of stakes (5, 7 or 9) makes perfect brace for weaving.

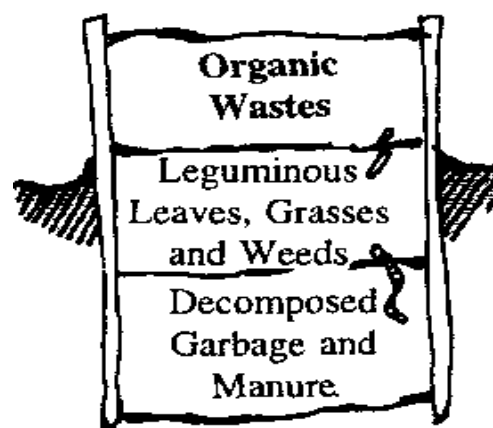
Weave the long strips of bamboo around the stakes to form a basket. Without bamboo strips, closely space the stakes (about 1 cm apart).



	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 28
			Revision # 00	

Addition of Organic Wastes

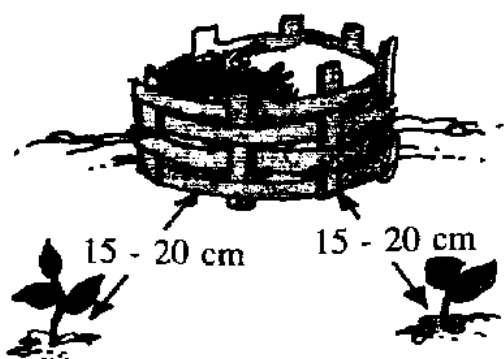
Place the most decomposed garbage and manure into the basket first. Place the undecomposed materials like leguminous leaves, grasses and weeds next. Fill to the brim with other organic wastes. Earthworms maybe added to speed up decomposition.



Planting and Care and Maintenance

Plant seeds or transplant seedlings around the basket. The distance from the basket should be 15 - 20 cm to prevent the decomposing materials from "burning" the plants.

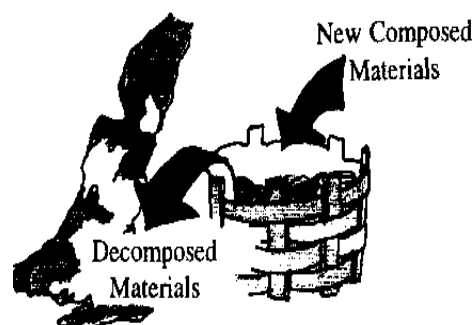
Water the seedlings while young. Eventually just water the basket. The plant roots will later move toward it.

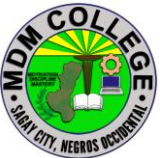


Incorporation of Decomposed Materials into the Soil

After harvesting, composts are already used up. Remove the decomposed materials from the basket and incorporate them into the soil while cultivating.

Add new composting materials to the basket for the next plants.



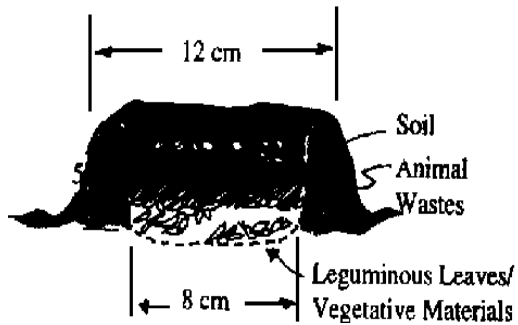
	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 29
			Revision # 00	

SELF CHECK 3.1-3

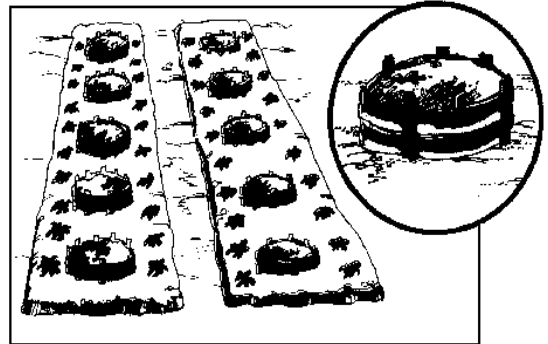
Prepare bed

Identify the following.

1.



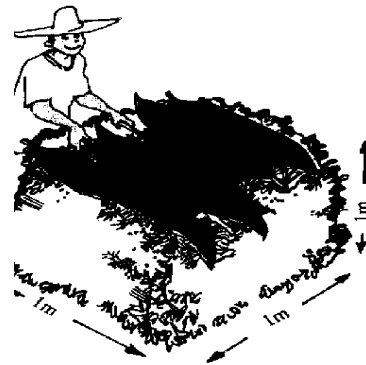
2.



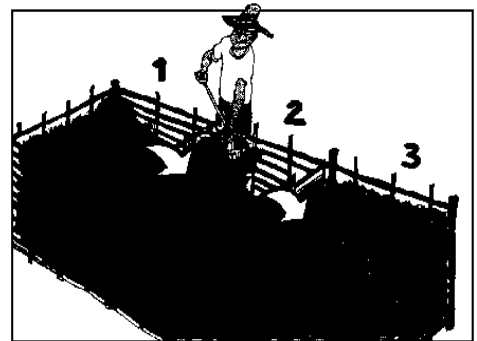
3.

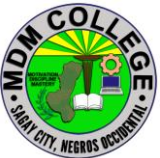


4.



5.

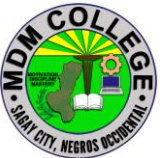


	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 30
			Revision # 00	

ANSWER KEY 3.1-3

Prepare bed

1. Cross-section of Composting Bed
2. Basket composting
3. Semi-sunken composting
4. The 14-day method of composting
5. Composting in triple-compost bin

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 31
			Revision # 00	

JOB SHEET 3.1-3

Prepare bed

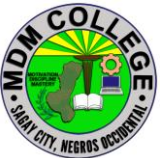
Title: Preparing composting bed
Performance Objective: At the end of this module the student should be able to Preparing composting bed.
Supplies/Materials: PPE, composting area, bamboo, shovel, raw material (animal manure, weeds, kitchens waste and crop residues).
Steps/Procedures: Wear the PPE Identify the raw materials Prepare bed for composting Check if the raw materials are mix Do housekeeping
Assessment Method: Demonstration

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 32
			Revision # 00	

PERFORMANCE CHECKLIST 3.1-3

Prepare bed

Performance Criteria	YES	NO
Did the trainee...		
1. Wear the PPE?		
2. Identify the raw materials		
3. Prepare bed for composting		
4. Check if the raw materials are mix		
5. Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 33
			Revision # 00	

INFORMATION SHEET 3.1-4

Gather materials

Learning Objective: After reading this information sheet, you should be able to gather the raw materials for composting.

Introduction

Composting is a great way to use the things in your refrigerator that are a little past their prime, which helps reduce food waste. You can also compost certain kinds of yard waste rather than send them to the dump. Collect these materials to start off your compost pile right:

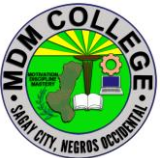
Step on gathering the materials

Step 1: Combine Green and Brown Materials

To make your own hot-compost heap, wait until you have enough materials to make a pile at least 3 feet deep. You are going to want to combine your wet, green items with your dry, brown items. "Brown" materials include dried plant materials; fallen leaves; shredded tree branches, cardboard, or newspaper; hay or straw; and wood shavings, which add carbon. "Green" materials include kitchen scraps and coffee grounds, animal manures (not from dogs or cats), and fresh plant and grass trimmings, which add nitrogen. For best results, start building your compost pile by mixing three parts brown with one part green materials. If your compost pile looks too wet and smells, add more brown items or aerate more often. If you see it looks extremely brown and dry, add green items and water to make it slightly moist.

Step 2: Water Your Pile

Sprinkle water over the pile regularly so it has the consistency of a damp sponge. Don't add too much water, otherwise, the microorganisms in your pile will become waterlogged and drown. If this happens, your pile will rot instead of compost.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 34
			Revision # 00	

Step 3: Stir Up Your Pile

During the growing season, you should provide the pile with oxygen by turning it once a week with a garden fork. The best time to turn the compost is when the center of the pile feels warm or when a thermometer reads between 130 and 150°F. Stirring up the pile will help it cook faster and prevents material from becoming matted down and developing an odor. At this point, the layers have served their purpose of creating equal amounts of green and brown materials throughout the pile, so stir thoroughly.

Step 4: Feed Your Garden

When the compost no longer gives off heat and becomes dry, brown, and crumbly, it's fully cooked and ready to feed to the garden. Add about 4 to 6 inches of compost to your flower beds and into your pots at the beginning of each planting season.

Some gardeners make what's known as compost tea with finished compost. This involves allowing fully formed compost to "steep" in water for several days, then straining it to use as a homemade liquid fertilizer.

Every gardener is different, so it's up to you to decide which composting method best fits your lifestyle. Fortunately, no matter which route you choose, compost is incredibly easy to make and environmentally friendly. Plus, it's a treat for your garden. With just a few kitchen scraps and some patience, you'll have the happiest garden possible.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 35
			Revision # 00	

SELF CHECK 3.1-4

Gather materials

True or False

____ 1. The 1st steps in gathering materials in compost is combination of brown and yellow.

____ 2. Fallen leaves, shredded tree branches, cardboard, or newspaper; hay or straw and wood shavings are brown materials.

____ 3. Green materials are the kitchen scraps and coffee grounds, animal manures and fresh plant.

____ 4. Water over the pile regularly so it has the consistency of a damp sponge.

____ 5. The best time to turn the compost is when the center of the pile feels warm or when a thermometer reads between 130 and 150°F.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 36
			Revision # 00	

ANSWER KEY 3.1-4

Gather material

1. False
2. True
3. True
4. True
5. true

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 37
			Revision # 00	

JOB SHEET 3.1-4

Gather material

Title: Gathering raw materials in composting
Performance Objective: At the end of this module you should be able to to gather the raw materials for composting.
Supplies/Materials: PPE, raw materials, water, spading fork.
Steps/Procedures: Wear the PPE. Gather the “green and brown “materials Put into the pile or compost pit Poured water regularly Mix the materials in pile using spading fork Do housekeeping.
Assessment method: Demonstration:

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 38
			Revision # 00	

PERFORMANCE CHECKLIST 3.4-1

Gather material

Performance Criteria	YES	NO
Did the trainee...		
Wear the PPE?		
Gather the “green and brown “materials?		
Put into the pile or compost pit?		
Poured water regularly?		
Mix the materials in pile using spading fork?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 39
			Revision # 00	

INFORMATION SHEET 3.1-5

Raw materials

Learning Objective: After reading this information sheet, you should be able to identify the raw materials uses for composting.

Introduction:

Some carbonaceous organic materials such as **rice straw, corn stalk, rice hull, and sawdust** are very useful in improving the physical and biological properties of soil, but they are very slow in releasing nutrients like nitrogen, phosphorus, and potassium.

Crop residues- Rice straw, rice hull, and other straws of graminaceous crops with abundant fibrous materials usually have a high C: N ratio, with a low nitrogen content but fairly high potassium and silica. Potassium and silica help improve the resistance of crops to disease and lodging, and fibrous materials provide an energy source for soil microorganisms as well as improve and condition soil physical properties. Crop residues are used as mulches to cover the surface of the soil and help maintain favorable soil moisture content and temperature as well as prevent the accumulation of salts or the multiplication of weeds on the soil surface. These materials can well be combined with swine or poultry manure that has a high nitrogen content to make better compost for crops (Lin et al.)

Green Manure-leguminous green manure crops are an important source of natural nitrogen. They fix nitrogen from the air and at flowering stage are usually incorporated into the soil, about ten days before planting the main crop. In extensively cropped areas, green manure crops are of great value to farmers since they reduce fertilizer costs. In intensively cropped areas, they may compete with the main crop for land. These green manure crops that have low C: N ratio (lower than 20 at vegetative stage) can be considered primary sources of nitrogen.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 40
			Revision # 00	

Animal Manure (Cattle, Goat, Swine, Chicken)

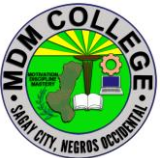
The nutrient content of swine manure is slightly higher than that of cattle manure, but with a higher copper content and lower content of fibrous material, discouraging repeated, long-term applications of this manure. It is best to dilute this manure by mixing it with rice hull, sawdust, rice straw, and similar fibrous materials and fermenting it before use.

The nutrient content of chicken manure is much higher than that of swine manure. However, its higher content of zinc and antibiotics and lower content of fibrous material discourage direct applications of fresh poultry manure to the soil. The best way to utilize this manure is to mix it with cattle and swine manure, rice straw, rice hull, sawdust, and other fibrous materials, and ferment it thoroughly before use.

Cattle manure has a reasonably high content of nitrogen, potassium, and fibrous materials. It is good animal manure because it does not have heavy metals and antibiotics in it. Repeated applications of this manure to the soil can be recommended, but phosphorus should be supplied from other sources to make up for its shortage in this manure. Nutrient content of goat manure is slightly higher than that of cattle manure.



Residues from oil extraction- compost consists mainly of sawdust and added with materials such as limestone and rice bran. Used mushroom compost has low potassium content as a result of leaching losses during mushroom culture, but the phosphorus, calcium, and C: N ratio and organic matter contents remain high and also, used mushroom compost has a high fibrous material content which improves soil physical properties and biological activity. However, the remnant mycelia in these materials may sometimes have a harmful effect on the roots of some crops. Therefore, it is recommended that used mushroom compost should be combined with a proper amount of high-nitrogen manure such as swine

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 41
			Revision # 00	

or poultry manure or oil extraction residues and be well fermented to kill the mycelia, before applying to the soil.

Residues from processing animal Product-The nutrient contents of animal residues differ greatly according to the type of residue. Animal blood, meat, horn, feet, wool, and feathers can all be used as a source of nitrogen fertilizer since they all have very high nitrogen content. Oyster shell and eggshell are good sources of calcium and bone meal can be a good source of phosphorus. However, all of them are very low in potassium. Fur should not be used in composting because of its high chromium content that can easily accumulate in the soil, causing toxicity in crops.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 42
			Revision # 00	

SELF CHECK 3.5-1

Raw Materials

True or False

_____ 1. Graminaceous crops materials usually have a high C: N ratio, with a low nitrogen content but fairly high potassium and silica, fibrous materials provide an energy source for soil microorganisms as well as improve and condition soil physical properties.

_____ 2. Green manure crops are an important source of natural nitrogen. They fix nitrogen from the air and at flowering stage are usually incorporated into the soil, about ten days before planting the main crop.

_____ 3. The best way to utilize this manure is to mix it with cattle and swine manure, rice straw, rice hull, sawdust, and other fibrous materials, and ferment it thoroughly before use.

_____ 4. Recommend that used mushroom compost should be combined with a proper amount of high-nitrogen manure such as swine or poultry manure or oil extraction residues and be well fermented to kill the mycelia, before applying to the soil.

_____ 5...Oyster shell and eggshell are good sources of calcium and bone meal can be a good source of phosphorus

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 43
			Revision # 00	

ANSWER KEY 3.1-5

Raw Materials

1. True
2. True
3. True
4. True
5. True

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 44
			Revision # 00	

JOB SHEET 3.5-1

Raw Materials

Title: Identifying raw materials for composting.

Performance Objective: At the end of this module you should be able to identify the raw materials uses for composting.

Supplies/Materials: PPE, dried leaves, rice straw, crop residues animal manure, oyster shell and egg shell.

Steps/Procedures:

Wear the PPE.

Identify the green and brown raw materials

utilize the manure animal manure

Do housekeeping.

Assessment method:

Demonstration:

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 45
			Revision # 00	

PERFORMANCE CHECKLIST 3.5-1

Raw Materials

Performance Criteria	YES	NO
Did the trainee...		
Wear the PPE?		
Identify the green and brown raw materials		
utilize the manure animal manure		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 46
			Revision # 00	

Learning Outcome #2 Compost and harvest fertilizer

Assessment Criteria:

Appropriate composting methods are applied based on production requirements
Compost is monitored based PNS indicators of fully decomposed fertilizer
Quality of harvest is checked based on PNS indicators of fully decomposed fertilizer
Processing of compost fertilizer are carried- out based on production requirement.
Record keeping is performed according to enterprise procedure.

Contents:

Composting methods
Decomposed fertilizer
Quality of harvest
Processing of compost fertilizer
Record keeping

Tools, Materials and Equipment and Facilities

Composting area
PPE (Boots, surgical masks, disposable gloves, overall

References:

<https://aseq-ehaq.ca/en/composting-101>

<https://directcompostsolutions.com/8-methods-composting/>

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 47
			Revision # 00	

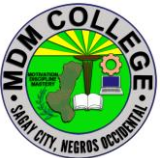
LEARNING EXPERIENCES/ACTIVITIES

LO # 2 Compost and harvest fertilizer

Learning Activities	Special Instructions
Read the attached Information Sheet 3.2-1 on Composting method	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.2-1 on Composting method	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Composting method Answer Key 3.2-1 on	
Perform Job 3.2-1 on Composting method	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance checklist on 3.2-1 on Composting method	
Read the attached Information Sheet 3.2-2 on Monitor Decomposition Process	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.2-2 on Monitor Decomposition Process	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.2-2 on Monitor Decomposition Process	
Perform Job Sheet 3.2-2 on Monitor Decomposition Process	Observe the demonstration of your trainer and check the procedures while performing the operation.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 48
			Revision # 00	

Evaluate performance using performance checklist 3.2-2 on Monitor Decomposition Process	
Read the attached Information Sheet 3.2-3 on Harvest Quality	You can ask the assistance of your trainer explain further the topics you <u>cannot understand</u>
Answer Self-Check 3.2-3 on Harvest Quality	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.2-3 on Harvest Quality	
Perform Job Sheet 3.2-3 on Harvest Quality	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance 3.2-3 on Harvest Quality	
Read the attached Information Sheet 3.2-4 on Processing of compost fertilizer	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.2-4 on Processing of compost fertilizer	Try to answer the Self-Check without looking at the information sheet
Compare your answer to Answer Key 3.2-4 on Processing of compost fertilizer	
Perform Job Sheet 3.2-4 on Processing of compost fertilizer	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance 3.2-4 on Processing of compost fertilizer	
Read the attached Information Sheet 3.2-5 on Record keeping	You can ask the assistance of your trainer explain further the topics you cannot understand
Answer Self-Check 3.2-5 on Record keeping	Try to answer the Self-Check without looking at the information sheet

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 49
			Revision # 00	

Compare your answer to Answer Key 3.2-5 on Record keeping	
Perform Job Sheet 3.2-5 on Record keeping	Observe the demonstration of your trainer and check the procedures while performing the operation.
Evaluate performance using performance 3.2-5 on Record keeping	

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	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 50
			Revision # 00	

INFORMATION SHEET 3.2-1

Composting method

Learning Objective: After reading this information sheet, you should be able to identify the different types of composting method.

Composting method of organic fertilizer Composting transforms raw organic residues into humus-like material through the activity of soil microorganisms. Mature compost stores well and is biologically stable, free of unpleasant odors, and easier to handle and less bulky than raw organic wastes. In agronomic and horticultural operations, compost can be used as a soil amendment, seed starter, mulch, container mix ingredient, or natural fertilizer, depending on its characteristics. Composting can also reduce or eliminate weed seeds and plant pathogens in organic residues

1. Sheet Composting “Sheet Composting, is also known as sheet mulching, can be a great way to add organic matter back into your soils. Essentially, this composting technique entails spreading thin layers of organic materials. (i. e. compost ingredients) on top of the soil surface. Some also consider this technique to be “composting in place.” Using green manure is another means of sheets composting. This technique is often used on the large scale; however, it can also be done successfully in your backyard”.

2. In – Vessel Composting “In – vessel composting is becoming more and more popular with large – scale compost producers. This method involves composting within an enclosed containment system, often a large cylindrical – shape container. The equipment involved in setting up an in – vessel composting system is typically quite expensive, and therefore usually limits it’s usage to industrial – sized composting operation. There are numerous benefits of in-vessel composting, such as an increased processing speed, year – round composting, and a highly controlled environment.

3. Anaerobic Composting “Anaerobic composting describes the biological breakdown of organic materials by living anaerobic organisms. This may not be the most odor- rific composting method, but it can be quite effective. Some of the benefits of composting anaerobically include the following: it is one of the most basic means of producing compost; it can be done on a small scale; and typically

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 51
			Revision # 00	

produces more humus per unit of starting materials than most other composting methods.”

4. Trench Composting “Trench composting involves digging holes in your garden soil and burying raw compost ingredients. Some people swear by this method, whereas others want nothing to do with it. Similar to anaerobic composting, this method of decomposition is quite simple; however, the materials tend to take longer to breakdown than when using other composting techniques.”

5. Bokashi Composting “Bokashi is a Japanese term meaning fermented organic matter. Therefore, bokashi composting describes the making of compost via fermentation. To achieve optimal results, your compost materials are inoculated with a microbial starter culture, and placed inside a sealed container. These starter cultures consist of several different species of microorganisms, all of which thrive in anaerobic conditions. One of the most popular microbial inoculants is called Effective Microorganisms of EM.”

6. Composting Barrels “Composting Barrels, or compost tumblers, are a great composting technique for backyard growers. They are self – contained, clean, and if big enough, can produce a fair amount of compost in a short period.” “You can buy composting barrels from a commercial supplier or you save your money and make one yourself.”

7. Vermicomposting Composting is the process of converting organic materials like leaves and animal manure into “humus” an organic matter through decomposition by the action of microbes and other organisms. Humus is rich in nutrients and contributes to soil texture and water retention.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 52
			Revision # 00	

SELF CHECK 3.2-1

Composting Method

True or False

____ 1. Trench composting involves digging holes in your garden soil and burying raw compost ingredients.

____ 2. Sheet Composting, is also as sheet mulching, can be a great way to add organic matter back into your soils.

____ 3. Humus is rich in nutrients and contributes to soil texture and water retention.

____ 4. Anaerobic composting describes the biological breakdown of organic materials by living anaerobic organisms.

____ 5. Aerobic composting describes the biological breakdown of organic materials by living anaerobic organisms.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 53
			Revision # 00	

ANSWER KEY 3.2-1

Method of composting

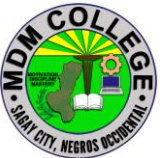
1. True
2. True
3. True
4. True
5. Falls

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 54
			Revision # 00	

JOB SHEET 3.2-1

Method of composting

Title: Identifying different types of composting method	
Performance Objective: At the end of this module the student should be able to Identify the different types of composting method.	
Supplies/Materials: composting area, vermi. Container, sacks and water	
Steps/Procedures: Wear PPE Check the area Identify the composting methods Apply vermi in the composting area Use sack to mulch Do not allow the air enter the compost Do housekeeping	
Assessment Method: Demonstration	

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 55
			Revision # 00	

PERFORMANCE CHECKLIST 3.2-1

Method of composting

Performance Criteria	YES	NO
Did the trainee...		
Wear PPE?		
Identify the composting methods?		
Apply vermi in the composting area?		
Use sack to mulch?		
Do not allow the air enter the compost?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 56
			Revision # 00	

INFORMATION SHEET 3.2-2

Monitor Decomposition Process

Learning Objective: After reading this information sheet, you should be able to monitor the decomposition process of organic fertilizer.

Introduction:

Composting is an aerobic method (meaning it requires air) of decomposing organic solid wastes. It can therefore be used to recycle organic material. The process involves decomposing organic material into a humus-like material, known as compost, which is a good fertilizer for plants.

Composting organisms require four equally important ingredients to work effectively:

- **Carbon** — for energy; the microbial oxidation of carbon produces the heat required for other parts of the composting process.¹ High carbon materials tend to be brown and dry.
- **Nitrogen** — to grow and reproduce more organisms to oxidize the carbon. High nitrogen materials tend to be green and wet. They can also include colorful fruits and vegetables.
- **Oxygen** — for oxidizing the carbon, the decomposition process. Aerobic bacteria needs oxygen levels >5% to perform the processes needed for composting.
- **Water** — in the right amounts to maintain activity without causing anaerobic conditions.

Certain ratios of these materials will allow microorganisms to work at a rate that will heat up the compost pile. Active management of the pile (e.g., turning) is needed to maintain sufficient oxygen and the right moisture level. The air/water balance is critical to maintaining high temperatures 130–160 °F (54–71 °C) until the materials are broken down.

Composting is most efficient with a carbon: nitrogen ratio of about 25:1. Hot container composting focuses on retaining heat in order to increase the decomposition rate thus producing compost more quickly. Rapid composting is favored by having a carbon: nitrogen ratio of ~30 or less. Above 30 the substrate

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 57
			Revision # 00	

is nitrogen starved. Below 15 it is likely to outgas a portion of nitrogen as ammonia.

Nearly all dead plant and animal materials have both carbon and nitrogen in different amounts. Fresh grass clippings have an average ratio of about 15:1 and dry autumn leaves about 50:1 depending upon species. Composting is an ongoing and dynamic process, adding new sources of carbon and nitrogen consistently as well as active management is important.

Organisms

Organisms can break down organic matter in compost if provided with the correct mixture of water, oxygen, carbon, and nitrogen. They fall into two broad categories: chemical decomposers which perform chemical processes on the organic waste, and physical decomposers which process the waste into smaller pieces through methods such as grinding, tearing, chewing, and digesting.

Chemical decomposers

Bacteria - The most abundant and important of all the **microorganisms** found in compost. Bacteria process carbon and nitrogen and excrete plant available nutrients such as nitrogen, phosphorus, and magnesium. Depending on the phase of composting, **mesophilic** or **thermophilic** bacteria may be the most prominent.

- **Mesophilic bacteria** get compost to the thermophilic stage through oxidation of organic material. Afterwards, they cure it which makes the fresh compost more bio-available for plants.
- **Thermophilic bacteria** do not reproduce and are not active between -5 to 25 °C yet are found throughout soil. They activate once the mesophilic bacteria have begun to breakdown organic matter and increase the temperature to their optimal range. They have been shown to enter soils via rainwater. They are present so broadly because of many factors including their spores being resilient. Thermophilic bacteria thrive from (40-60 °C), and only large-scale composting - such as windrow composting - operations generally exceed (60-65 °C), beyond which point many beneficial microorganisms will die.
- **Actinobacteria** are needed to break down paper products such as newspaper, bark, etc and other large molecules such as lignin and cellulose that are more difficult to decompose. The "pleasant earthy smell of compost" is attributed to actinobacteria. They make carbon, ammonia, and nitrogen nutrients available to plants.
- Fungi such as mold and yeast help break down materials that bacteria cannot, especially cellulose and lignin in woody material.
- **Protozoa** - contribute to biodegradation of organic matter as well as consuming non-active bacteria, fungi, and micro-organic particulates.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 58
			Revision # 00	

Physical decomposers Ants - create nests, making the soil more porous and transporting nutrients to different areas of the compost. Beetles - grubs feed on decaying vegetables.

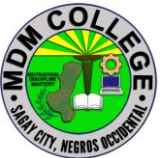
- **Earthworms** - ingest partly composted material and excrete worm castings, making nitrogen, calcium, phosphorus, and magnesium available to plants. The tunnels they create as they move through the compost also increase aeration and drainage.
- **Flies** - feed on almost all organic material and input bacteria into the compost. Their population is kept in check by mites and the thermophilic temperatures that are unsuitable for fly larvae.
- **Millipedes** - break down plant material.
- **Rotifers** - feed on plant particles.
- **Snails and Slugs** - feed on living or fresh plant material. They should be removed from compost before use as they can damage plants and crops.
- **Sow Bugs** - feed on rotting wood, and decaying vegetation.
- **Springtails** - feed on fungi, mold, and decomposing plants.

Phases of composting

1. Mesophilic phase: An initial, mesophilic phase, in which the decomposition is carried out under moderate temperatures by mesophilic microorganisms.

2. Thermophilic phase: As the temperature rises, a second, thermophilic phase starts, in which various thermophilic bacteria carry out the decomposition under higher temperatures (50 to 60 °C (122 to 140 °F).)

3. Maturation phase: As the supply of high-energy compounds dwindles, the temperature starts to decrease, and the mesophilic bacteria once again predominate in the maturation phase

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 59
			Revision # 00	

SELF CHECK 3.2-2

Monitor Decomposition Process

Identification

1. For energy the microbial oxidation of carbon produces the heat required for other parts of the composting process. High carbon materials tend to be brown and dry.
2. Grow and reproduce more organisms to oxidize the carbon. High nitrogen materials tend to be green and wet. They can also include colorful fruits and vegetables.
3. For oxidizing the carbon, the decomposition process. Aerobic bacteria need oxygen levels >5% to perform the processes needed for composting.
4. The right amounts to maintain activity without causing anaerobic conditions.
5. can break down organic matter in compost if provided with the correct mixture of water, oxygen, carbon, and nitrogen

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 60
			Revision # 00	

ANSWER KEY 3.2-2

Monitor Decomposition Process

1. Water
2. Oxygen
3. Nitrogen
4. Carbon
5. Organisms

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 61
			Revision # 00	

JOB SHEET 3.2-2

Monitor Decomposition Process

Title: Monitoring Decomposition process
Performance Objective: At the end of this module the student should be able to monitor decomposition process.
Supplies/Materials: PPE, composting area.
Steps/Procedures: Wear the PPE. Check the compost pile Monitor the raw materials if undergo decomposition process
Assessment Method: Demonstration

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 62
			Revision # 00	

PERFORMANCE CHECKLIST 3.2-2

Decomposition process

Performance Criteria	YES	NO
Did the trainee...		
Wear the PPE?		
Check the compost pile?		
Monitor the raw materials if undergo decomposition process?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 63
			Revision # 00	

INFORMATION SHEET 3.2-3

Harvest Quality

Learning Objective: After reading this information sheet, you should be able to identify the quality of a good harvest.

Introduction:

Compost is ready or finished when it looks, feels and smells like rich, dark earth rather than rotting vegetables. In other words, it should be dark brown, crumbly and smell like earth. The decomposition of kitchen and yard waste into finished compost can take anywhere from a few months to a few years depending on how well you manage the balancing the food (greens and browns), water, and air.

How to Harvest

How you harvest your mature compost will be dependent on what style of compost bin you use. If you have a single compost bin, you will likely have to remove the top portion of the pile to get to the material that is ready for harvesting (it falls to the bottom). If you have several bins, or a multi-bin system, you can use one section or bin to let compost fully mature, and add new materials to the other(s). That way you can simply empty out the entire bin when its contents are mature! With a single bin you will either have to:

- Move your bin over adjacent to its current location and transfer the unfinished material back into the bin. You can then harvest the remaining material.
- Remove the top portion of the pile. Once you have harvested the finished compost from below, you can then shovel the contents back into the bin and start adding new material on top.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 64
			Revision # 00	

When to Harvest

Generally compost is ready to be harvested when the finished product is a rich dark brown color, smells like earth, and crumbles in your hand. Some signs that it may not be ready include:

- Recognizable food content still visible
- The pile is still warm. This means that it is still working.
- There are still lots of large lumps in the pile.

If any of the above are noticeable, give it a few more weeks and check it again. If it looks ready but you still have twigs and other hard materials like fruit pits it can still be considered finished. You may however need to screen it before using it, depending on what you intend to use it for. Screened compost makes a great top dressing for your lawn.

Many people ask us is it better to harvest compost in the spring or the fall? Generally, we recommend harvesting in the fall because if compost is used before it has fully matured, the microorganisms may rob the soil and plants of nitrogen in order to finish the process. Spreading your compost in the fall will ensure that by spring any issues related to compost maturity will be eliminated



How to use finished compost

Finished compost can be used both outdoors and indoors as a natural fertilizer for plants. Mix it in the soil before planting, or use it to top dress your indoor and outdoor plants, including your lawn and shrubs.

Compost can also be used to make a rich liquid fertilizer called compost tea which can not only be used to provide nutrients for your plants, it can also suppress leaf disease when sprayed directly on the plant leaves.



	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 65
			Revision # 00	

SELF CHECK 3.2-3

Harvest Quality

TRUE and FALSE

1. Compost is ready to be harvested when the finished product is a rich dark red color.
2. Compost is ready to be harvested when the finished product is a rich smell like globe.
3. Compost is ready to be harvested when the finished product is a rich crumble in hands.
4. Compost can also be used to make a rich liquid fertilizer called compost tea.
5. Compost fertilizer can be used to provide nutrients to the plants.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 66
			Revision # 00	

ANSWER KEY 3.2-3

Harvest Quality

1. Falls
2. Falls
3. True
4. True
5. True

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 67
			Revision # 00	

JOB SHEET 3.2-3

Quality Harvest

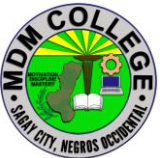
Title: Harvesting time
Performance Objective: At the end of this module the student should be able to identify the good quality time of harvest.
Supplies/Materials: PPE, Compost.
Steps/Procedures: Wear the PPE. Identify the good quality time of harvest Select the compost ready to harvest Collect the compost Do housekeeping
Assessment Method: Demonstration

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 68
			Revision # 00	

PERFORMANCE CHECKLIST 3.2-3

Quality Harvest

Performance Criteria	YES	NO
Did the trainee...		
1. Wear the PPE?		
2 Identify the good quality time to harvest?		
3. Select the compost ready to harvest?		
4. Collect the compost?		
5. Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 69
			Revision # 00	

INFORMATION SHEET 3.2-4

Processing of Compost Fertilizer

Learning Objective: After reading this information sheet, you should be able to identify processes of composting fertilizer.

Introduction:

Composting process it is important to understand the process of composting. It is primarily a microbiological process of decomposing organic substances, which results in a product that is stable, pathogen-free, and contains readily available nutrients when applied to the soil. In the early stages of composting, the first to decompose are the proteins, sugars and starches then the cellulose and hemicelluloses, and lastly the lignin.

1. Collect substrates. In collecting substrate, a hand tractor with trailer may be needed. Collect crop residues separately from animal manures and rice hull ash or carbonized rice hull. Bring them to the composting area. Composting area should have a shade to avoid direct rainfall that could leach out nutrients from the decomposing organic materials.

2. Mix the materials at a ratio of 3:1:1(3 solid wastes, 1 farm manure, and 1 rice hull ash or CRH.

3. Moisten the materials then shred to enhance decomposition. Shredding will reduce the size of the materials for easier attack of microorganisms.

4. In the absence of shredding machine, use microbial enhancers, which are available in the market. Follow the procedure listed in the packet of the microbial enhancers.

5. Pile the shredded materials and cover with used plastic to minimize evolution of greenhouse gases and conserve moisture of the pile. Be sure that the plastic cover is perforated to allow aeration in the pile.

6. Monitor the MC and temperature of the pile every 2 days. In monitoring the pile temperature, insert soil thermometer in each of the sides of the pile as well as on top of the pile. Record the temperature. In the absence of soil thermometer, insert your hand on the sides of the pile and feel the heat of the pile. Presence of

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 70
			Revision # 00	

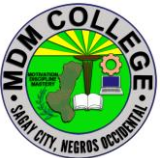
heat indicates active composting process. If the pile is dry, add water. Be sure that enough water is added to the pile to maintain moist condition. Get sample of the materials and hold it in your hands. If water drips from your hand, there is much water in your pile.

7. After 2 weeks, open the pile and turn it. Turning can be done by spading the end of the pile and turning it back. Afterward, ensure that compost material is moist and then cover with plastic.

8. After another 2 weeks, the compost is more or less mature already. Mature compost is indicated by black/dark colored material absence of foul odor, no heat dissipated, and unrecognizable original compost materials.

9. Air-dry the compost under a shaded place to avoid direct impact of rainfall. Do not over dry the compost. Maintain compost MC at least approximately 30%. Get a handful sample of the compost and squeeze in your palm. If no water drips and you can form a ball, then MC of the compost is approximately 30%.

10. Shred the compost to have a uniform size of the material, sieve, and place in a sack with plastic sack inside. An ordinary sack will contain 50kg of the compost material. Sew the sack and store in a cool dry place. Below is the average nutrient composition of composted agricultural wastes.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 71
			Revision # 00	

SELF CHECK 3.2-4

Processing of Compost Fertilizer

True or Falls

____ 1. The process of composting. It is primarily a microbiological process of decomposing organic substances, which results in a product that is stable, pathogen-free, and contains readily available nutrients when applied to the soil.

____ 2. Moisten the materials then shred to enhance decomposition. Shredding will reduce the size of the materials for easier attack of microorganisms.

____ 3. Pile the shredded materials and cover with used plastic to minimize evolution of greenhouse gases and conserve moisture of the pile. Be sure that the plastic cover is perforated to allow aeration in the pile.

____ 4. After 2 weeks, open the pile and turn it. Turning can be done by spading the end of the pile and turning it back. Afterward, ensure that compost material is moist and then cover with plastic.

____ 5. Air-dry the compost under a shaded place to avoid direct impact of rainfall. Do not over dry the compost. Maintain compost MC at least approximately 30%. Get a handful sample of the compost and squeeze in your palm

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 72
			Revision # 00	

ANSWER KEY 3.2-4

Processing of Compost Fertilizer

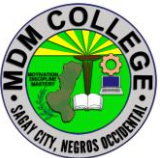
1. True
2. True
3. True
4. True
5. True

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 73
			Revision # 00	

JOB SHEET 3.2-4

Processing of Compost Fertilizer

Title: Composting Process	
Performance Objective: At the end of this module you should be able to appreciate the importance of composting process.	
Supplies/Materials: PPE, composting area, substrates, plastic and water	
Steps/Procedures: Wear the PPE. Check the composting area Collect the substrate Put in the pile Check the microorganism attack the substrate Monitor the pile regularly Do housekeeping.	
Assessment method: Demonstration:	

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 74
			Revision # 00	

PERFORMANCE CHECKLIST 3.2-4

Processing of Compost Fertilizer

Performance Criteria	YES	NO
Did the trainee...		
Wear the PPE?		
Check the composting area?		
Collect the substrate?		
Put the substrate inside the pile?		
Check the microorganism attack the substrate?		
Monitor the pile regularly?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 75
			Revision # 00	

INFORMATION SHEET 3.2-5

Record Keeping

Learning Objective: After reading this information sheet, you should be able to perform record keeping.

Introduction:

Keeping accurate and detailed records is an important aspect of managing a compost system. Accurate records provide crucial feedback about how the system is working and information upon which to base decisions or take action. Everyone on the compost team should have an understanding of why these records are important and how to make accurate measurements. Good record keeping also ensures that if site issues develop or the compost team encounters neighbor complaints, the issues and any actions taken are documented.

The ability to make good compost starts with a good recipe and with your ability to monitor the compost and respond to changes as they occur. Many operators suggest there is "an art to composting". Good records can help you to improve your compost quality. It is important to keep records of the feedstock materials being composted, C: N ratios and approximate moisture content at the start of the composting process.

Records should also include when the compost was turned and the conditions of the compost at that time. Temperatures need to be monitored regularly. Some standard operating procedures require daily temperature recording during the active phase, to ensure that the required temperatures have been achieved. It is recommended that the thermometer has a one meter (or three-foot stem) with a 0-80 °C temperature range, and be inserted into the center of the pile in several locations at each monitoring event. Make notes of moisture condition and any odors. Foul odors may indicate anaerobic conditions or lack of oxygen. Ammonia smells may indicate high nitrogen content (C: N imbalance) and a need for more carbon material.

Additional records and sampling may be required when composting permits are required, such as when off-farm materials are added, or when selling compost as a soil amendment or fertilizers.

Record keeping provides valuable information concerning what worked and what did not and possibly, the reasons why something did not turn out as planned. As we all know, the best production methods and hybrids can fail due to weather

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 76
			Revision # 00	

conditions. Keeping daily records of precipitation and high and low temperatures is easy to accomplish and you can have a fairly accurate weather station for as little. A high/low thermometer, rain gauge, something to record them on, and about ten minutes a day is all that is needed.

One set of accurate records should be enough to satisfy all of your needs. If you keep good records of production, expenses, income, and weather, you should have all of the required information for any purpose, from crop insurance to lenders to your own needs you should be able to make informed conclusions regarding the past and to plan for the future.

Good records lead to better decisions and hopefully higher returns.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 77
			Revision # 00	

SELF CHECK 3.2-5

Record Keeping

1-5. Give 5 reason why record keeping are important in composting.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 78
			Revision # 00	

ANSWER KEY 3.2-5

Record Keeping

1. Accurate records provide crucial feedback about how the system is working and information upon which to base decisions or take action.
2. Good record keeping also ensures that if site issues develop or the compost team encounters neighbor complaints, the issues and any actions taken are documented.
3. Important to keep records of the feedstock materials being composted, C: N ratios and approximate moisture content at the start of the composting process.
4. Records should also include when the compost was turned and the conditions of the compost at that time. Temperatures need to be monitored regularly. Some standard operating procedures require daily temperature recording during the active phase, to ensure that the required temperatures have been achieved.
5. Accurate records should be enough to satisfy all of your needs. If you keep good records of production, expenses, income, and weather, you should have all of the required information for any purpose, from crop insurance to lenders to your own needs you should be able to make informed conclusions regarding the past and to plan for the future.

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 79
			Revision # 00	

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 80
			Revision # 00	

JOB SHEET 3.2-5

Record Keeping

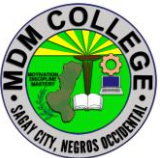
Title: Record Keeping
Performance Objective: At the end of this module you should be able to keep record.
Supplies/Materials: PPE, composting area, logbook, ball pens
Steps/Procedures: Wear the PPE. Prepare the logbook Check the composting pile Record the inputs Records the output of composting Do housekeeping.
Assessment method: Demonstration:

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 81
			Revision # 00	

PERFORMANCE CHECKLIST 3.2-5

Record Keeping

Performance Criteria	YES	NO
Did the trainee..		
Wear the PPE?		
Prepare the logbook?		
Check the composting pile?		
Record the inputs?		
Records the output of composting?		
Do housekeeping?		

	Organic Agriculture NC II Produce Organic Fertilizer	September 1, 2019	Document No.	
		Nonie Falcunit	Issued by:	Page 82
			Revision # 00	